

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

50 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

50 KWP GRID-TIED SOLAR PLANT

Net investment ₹21.90 L after input tax credit, generating 76,650 units in year one against an energy charge of ₹10.15 per unit.

LT CATEGORY II — COMMERCIAL

30-YEAR ANALYSIS

PAYBACK

3.2 yrs

Post-tax, undiscounted

PROJECT IRR

31.3%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.52

per kWh, against ₹10.15 from the grid

NET PRESENT VALUE

₹50.14 L

Discounted at 10%

PLANT AND GENERATION

Plant capacity	50 kWp
Year 1 generation	76,650 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	LT Category II — Commercial (LT supply)
Energy charge offset	₹10.15 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹7.21 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹46.00 / Wp (incl. GST)
Project price incl. GST	₹23.00 L
GST at 5% — recovered as ITC	₹1.10 L
Net capitalised cost	₹21.90 L
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹5.51 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	21,90,476	8,76,190	13,14,286	2,20,519
2	13,14,285	5,25,714	7,88,571	1,32,312
3	7,88,572	3,15,429	4,73,143	79,387
4	4,73,143	1,89,257	2,83,886	47,632
5	2,83,885	1,13,554	1,70,331	28,579
6	1,70,332	68,133	1,02,199	17,148
7	1,02,199	40,880	61,319	10,289
8	61,320	24,528	36,792	6,173
9	36,792	14,717	22,075	3,704
10	22,075	8,830	13,245	2,222
11	13,245	5,298	7,947	1,333
12	7,947	3,179	4,768	800
Tax shield, years 1-12				5,50,099

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

50 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	76,650	7,21,277	—	8,76,190	(38,989)	7,60,265	7,60,265
2	76,267	7,46,377	30,000	5,25,714	47,986	6,68,391	14,28,656
3	75,885	7,72,351	30,000	3,15,429	1,07,448	6,34,903	20,63,559
4	75,506	7,99,229	30,000	1,89,257	1,45,967	6,23,261	26,86,820
5	75,128	8,27,042	30,000	1,13,554	1,72,020	6,25,022	33,11,842
6	74,753	8,55,823	30,000	68,133	1,90,695	6,35,127	39,46,969
7	74,379	8,85,605	30,000	40,880	2,05,050	6,50,555	45,97,524
8	74,007	9,16,425	30,000	24,528	2,16,922	6,69,502	52,67,026
9	73,637	9,48,316	30,000	14,717	2,27,418	6,90,898	59,57,924
10	73,269	9,81,318	30,000	8,830	2,37,205	7,14,112	66,72,036
11	72,903	10,15,467	30,000	5,298	2,46,689	7,38,778	74,10,814
12	72,538	10,50,806	30,000	3,179	2,56,116	7,64,689	81,75,503
13	72,175	10,87,374	30,000	1,907	2,65,640	7,91,734	89,67,237
14	71,815	11,25,214	30,000	1,144	2,75,356	8,19,859	97,87,096
15	71,455	11,64,372	30,000	687	2,85,326	8,49,046	1,06,36,142
16	71,098	12,04,892	30,000	412	2,95,593	8,79,299	1,15,15,441
17	70,743	12,46,822	30,000	247	3,06,188	9,10,635	1,24,26,076
18	70,389	12,90,212	30,000	148	3,17,133	9,43,079	1,33,69,155
19	70,037	13,35,111	30,000	89	3,28,448	9,76,663	1,43,45,818
20	69,687	13,81,573	30,000	53	3,40,150	10,11,422	1,53,57,240
21	69,338	14,29,651	30,000	32	3,52,256	10,47,395	1,64,04,635
22	68,992	14,79,403	30,000	19	3,64,781	10,84,622	1,74,89,257
23	68,647	15,30,887	30,000	12	3,77,740	11,23,146	1,86,12,403
24	68,304	15,84,161	30,000	7	3,91,150	11,63,012	1,97,75,415
25	67,962	16,39,290	30,000	4	4,05,025	12,04,265	2,09,79,680
26	67,622	16,96,338	30,000	2	4,19,383	12,46,954	2,22,26,634
27	67,284	17,55,370	30,000	1	4,34,241	12,91,129	2,35,17,763
28	66,948	18,16,457	30,000	1	4,49,615	13,36,842	2,48,54,605
29	66,613	18,79,670	30,000	1	4,65,525	13,84,145	2,62,38,750
30	66,280	19,45,082	30,000	0	4,81,988	14,33,094	2,76,71,844
Total	21,40,311	3,71,11,915	8,70,000	21,90,475	85,70,065	2,76,71,844	2,76,71,844

RETURN METRICS

Simple payback	3.2 yrs
Discounted payback	4.1 yrs
Project IRR, unlevered	31.3%
Net present value at 10%	₹50.14 L

OVER THE FULL TERM

Levelised cost of energy	₹3.52 / kWh
Cumulative return, 30 years	1163%
Total net cash flow, after tax	₹2.77 Cr
30-year generation	21,40,310 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.52
Grid energy charge today		₹10.15

Every unit the plant generates costs ₹3.52 across its life against ₹10.15 bought from APSPDCL today — and the grid figure reaches ₹31.65 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 21,40,310 units, avoiding 1,755.1 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** LT Category II — Commercial at ₹10.15/kWh. Marginal rate, above 500 units per month. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹23.00 L.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 50 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.